Optimal Control and Reinforcement Learning
Optimality conditions for Optimal Control

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Discrete-time control system and notation (Refresh)

We consider nonlinear, discrete-time systems described by

$$x_{t+1} = f_t(x_t, u_t) \qquad t \in \mathbb{N}_0 \tag{1}$$

where $x_t \in \mathbb{R}^n$ and $u_t \in \mathbb{R}^m$ are the state and the input of the system at time t.

Notation

We use $\mathbf{x} \in \mathbb{R}^{nT}$ and $\mathbf{u} \in \mathbb{R}^{mT}$ to denote, respectively, the stack of the states x_t for all $t \in 1, \dots, T$ and the inputs u_t for all $t \in 0, \dots, T-1$, that is

$$\mathbf{x} := \mathrm{col}(x_1, \dots, x_T)$$
$$\mathbf{u} := \mathrm{col}(u_0, \dots, u_{T-1}).$$

where $\mathrm{col}(x_1, \dots, x_T)$ is the (column) vector with components $x_1, \dots, x_T$.

System trajectories (Refresh)

We recall the definition of trajectory of a (possibly) nonlinear, discrete-time control system.

Definition A pair $(\bar{\mathbf{x}}, \bar{\mathbf{u}}) \in \mathbb{R}^{nT} \times \mathbb{R}^{mT}$ is called a *trajectory* of system (1) if $\bar{x}_{t+1} = f_t(\bar{x}_t, \bar{u}_t)$ for all $t \in 0, \dots, T-1$. In particular, $\bar{\mathbf{x}}$ is the state trajectory, while $\bar{\mathbf{u}}$ is the input trajectory. □

Remark It will be sometimes useful to refer to a generic pair $(\boldsymbol{\alpha}, \boldsymbol{\mu}) \in \mathbb{R}^{nT} \times \mathbb{R}^{mT}$ with $\boldsymbol{\alpha} := \mathrm{col}(\alpha_1, \dots, \alpha_T)$ and $\boldsymbol{\mu} := \mathrm{col}(\mu_0, \dots, \mu_{T-1})$ as a state-input *curve*. Notice that a curve $(\boldsymbol{\alpha}, \boldsymbol{\mu})$ is not necessarily a trajectory, i.e., it does not necessarily satisfy the dynamics (1). □

Dynamics as equality constraints

Let us rewrite the nonlinear dynamics (1) as an implicit equality constraint $h : \mathbb{R}^{nT} \times \mathbb{R}^{mT} \to \mathbb{R}^{nT}$

$$h(\mathbf{x}, \mathbf{u}) := \begin{bmatrix} f_0(x_0, u_0) - x_1 \\ \vdots \\ f_{T-1}(x_{T-1}, u_{T-1}) - x_T \end{bmatrix}, \tag{2}$$

so that a curve $(\bar{\mathbf{x}}, \bar{\mathbf{u}})$ is a trajectory of (1) if it satisfies the (possibly nonlinear) equality constraint

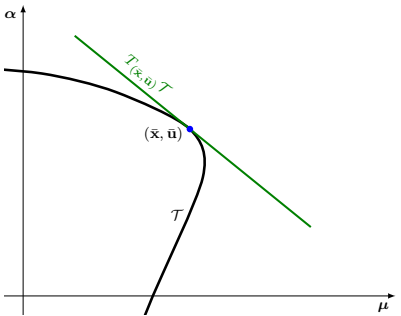
$$h(\bar{\mathbf{x}}, \bar{\mathbf{u}}) = 0.$$

Remark Recall that a nonlinear equality constraint is a nonconvex constraint. □

System trajectories and trajectory manifold

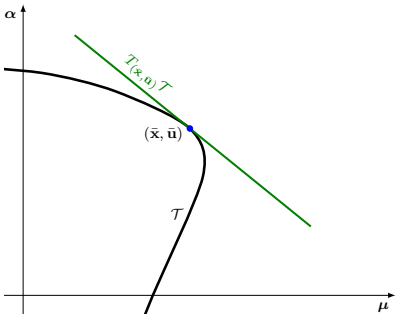
We can now define the trajectory manifold $\mathcal{T} \subset \mathbb{R}^{nT} \times \mathbb{R}^{mT}$ of (1) as

$$\mathcal{T} := \{(\mathbf{x}, \mathbf{u}) \in \mathbb{R}^{nT} \times \mathbb{R}^{mT} \mid h(\mathbf{x}, \mathbf{u}) = 0\} = \{(\mathbf{x}, \mathbf{u}) \in \mathbb{R}^{nT} \times \mathbb{R}^{mT} \mid x_{t+1} = f_t(x_t, u_t), t = 0, \dots, T-1\}. \tag{3}$$



System linearization and tangent hyperplane

Let $(\bar{\mathbf{x}}, \bar{\mathbf{u}}) \in \mathcal{T}$ be a trajectory of the system, i.e., a point on the trajectory manifold $\mathcal{T}$.



The tangent space to $\mathcal{T}$ at a given trajectory (point) $(\bar{\mathbf{x}}, \bar{\mathbf{u}})$, denoted as $T_{(\bar{\mathbf{x}},\bar{\mathbf{u}})}\mathcal{T}$, is the set of trajectories satisfying the linearization of $x_{t+1} = f_t(x_t, u_t)$ about the trajectory $(\bar{\mathbf{x}}, \bar{\mathbf{u}})$.

System linearization and tangent hyperplane

That is, $T_{(\bar{\mathbf{x}},\bar{\mathbf{u}})}\mathcal{T} = \{(\boldsymbol{\Delta}\mathbf{x}, \boldsymbol{\Delta}\mathbf{u}) \in \mathbb{R}^{nT} \times \mathbb{R}^{mT} \mid \nabla_1 h(\mathbf{x},\mathbf{u})^\top \boldsymbol{\Delta}\mathbf{x} + \nabla_2 h(\mathbf{x},\mathbf{u})^\top \boldsymbol{\Delta}\mathbf{u} = 0\}$ is the set of trajectories $(\boldsymbol{\Delta}\mathbf{x}, \boldsymbol{\Delta}\mathbf{u})$ of

$$\Delta x_{t+1} = A_t \Delta x_t + B_t \Delta u_t,$$

with

$$A_t = \nabla_1 f_t(\bar{x}_t, \bar{u}_t)^\top$$
$$B_t = \nabla_2 f_t(\bar{x}_t, \bar{u}_t)^\top.$$

Remark Recall that for a vector function $f : \mathbb{R}^{n+m} \to \mathbb{R}^n$, ∇f is the transpose of the Jacobian matrix.

Unconstrained optimal control problem (discrete-time)

We look for a solution of the following discrete-time optimal control problem

$$\min_{\mathbf{x}\in\mathbb{R}^n, \mathbf{u}\in\mathbb{R}^m} \quad \sum_{t=0}^{T-1} \ell_t(x_t, u_t) + \ell_T(x_T) \tag{4a}$$

$$\text{subj.to } x_{t+1} = f_t(x_t, u_t), \quad t \in 0, \dots, T-1 \tag{4b}$$

with given initial condition $x_0 = x^0 \in \mathbb{R}^n$, where $\ell_t : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is the so-called stage cost while $\ell_T : \mathbb{R}^n \to \mathbb{R}$ is the terminal cost.

Remark An optimal control problem constrained only through the dynamics (without boundary and path constraints) is named unconstrained optimal control. This notation will be clarified in the following.

Unconstrained optimal control problem (discrete-time)

We look for a solution of the following discrete-time optimal control problem

$$\min_{\mathbf{x}\in\mathbb{R}^n, \mathbf{u}\in\mathbb{R}^m} \quad \sum_{t=0}^{T-1} \ell_t(x_t, u_t) + \ell_T(x_T) \tag{4a}$$

$$\text{subj.to } x_{t+1} = f_t(x_t, u_t), \quad t \in 0, \dots, T-1 \tag{4b}$$

with given initial condition $x_0 = x^0 \in \mathbb{R}^n$, where $\ell_t : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is the so-called stage cost while $\ell_T : \mathbb{R}^n \to \mathbb{R}$ is the terminal cost.

From now on, we will use the following assumption.
Assumption Functions $\ell_t(\cdot,\cdot)$, $\ell_T(\cdot)$, $f_t(\cdot,\cdot)$ are twice continuously differentiable, i.e., they are class $\mathcal{C}^2$.

Unconstrained optimal control problem (compact notation)

Consider the discrete-time system $x_{t+1} = f_t(x_t, u_t)$. We can introduce the compact notation

$$\mathbf{x} := \begin{bmatrix} x_1 \\ \vdots \\ x_T \end{bmatrix} \qquad \mathbf{u} := \begin{bmatrix} u_0 \\ \vdots \\ u_{T-1} \end{bmatrix}$$

which let us write compactly the cost function as

$$\ell(\mathbf{x}, \mathbf{u}) := \sum_{t=0}^{T-1} \ell_t(x_t, u_t) + \ell_T(x_T).$$

and the equality constraint represented by the dynamics

$$h(\mathbf{x}, \mathbf{u}) := \begin{bmatrix} f_0(x_0, u_0) - x_1 \\ \vdots \\ f_{T-1}(x_{T-1}, u_{T-1}) - x_T \end{bmatrix}$$

Optimal control as equality constrained optimization

In light of this compact notation, we can rewrite the optimal control problem as

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^{nT},\mathbf{u}\in\mathbb{R}^{mT}} \quad & \ell(\mathbf{x},\mathbf{u}) \\ \text{subj.to} \quad & h(\mathbf{x},\mathbf{u}) = 0 \end{aligned}$$

where $\ell : \mathbb{R}^{nT} \times \mathbb{R}^{mT} \to \mathbb{R}$ and $h : \mathbb{R}^{nT} \times \mathbb{R}^{mT} \to \mathbb{R}^{nT}$.

This is a constrained nonlinear optimization problem with decision variable $(\mathbf{x}, \mathbf{u})$.

Optimality conditions (Refresh from "Optimization" part)

Let z^* be a regular local minimum of the equality-constrained minimization problem:

$$\begin{aligned} \min_{z \in \mathbb{R}^n} \quad & \ell(z) \\ \text{subj.to} \quad & h(z) = 0 \end{aligned}$$

where $\ell : \mathbb{R}^n \to \mathbb{R}$ and $h : \mathbb{R}^n \to \mathbb{R}^n$.
Then

$$\begin{aligned} \nabla \ell(z^*) + \nabla h(z^*) \lambda^* &= 0 \\ h(z^*) &= 0 \end{aligned}$$

which can be written as

$$\nabla \mathcal{L}(z^*, \lambda^*) = \begin{bmatrix} \nabla \ell(z^*) + \nabla h(z^*) \lambda^* \\ h(z^*) \end{bmatrix} = 0.$$

Remark If you have doubts go back to Part "Equality and Inequality Constrained Optimization".

KKT conditions for unconstrained optimal control

KKT conditions for unconstrained optimal control

The Lagrangian function has the form

$$\begin{aligned}\mathcal{L}(\mathbf{x},\mathbf{u},\boldsymbol{\lambda}) &= \ell(\mathbf{x},\mathbf{u}) + \boldsymbol{\lambda}^\top h(\mathbf{x},\mathbf{u}) \\ &= \underbrace{\sum_{t=0}^{T-1} \ell_t(x_t,u_t) + \ell_T(x_T)}_{\ell(\mathbf{x},\mathbf{u})} + \underbrace{\sum_{t=0}^{T-1} \lambda_{t+1}^\top (f_t(x_t,u_t) - x_{t+1})}_{\boldsymbol{\lambda}^\top h(\mathbf{x},\mathbf{u})} \\ &= \sum_{t=0}^{T-1} \Big(\ell_t(x_t,u_t) + \lambda_{t+1}^\top (f_t(x_t,u_t) - x_{t+1})\Big) + \ell_T(x_T) \\ &= \sum_{t=0}^{T} \mathcal{L}_t(x_t,u_t,\boldsymbol{\lambda})\end{aligned}$$

where $\boldsymbol{\lambda} \in \mathbb{R}^{nT}$ and

$$\begin{aligned}\mathcal{L}_0(x_0,u_0,\boldsymbol{\lambda}) &= \ell_0(x_0,u_0) + \lambda_1^\top f_0(x_0,u_0) \\ \mathcal{L}_t(x_t,u_t,\boldsymbol{\lambda}) &= \ell_t(x_t,u_t) + \lambda_{t+1}^\top f_t(x_t,u_t) - \lambda_t x_t \qquad t = 1,\dots,T-1 \\ \mathcal{L}_T(x_T,\boldsymbol{\lambda}) &= \ell_T(x_T) - \lambda_T^\top x_T\end{aligned}$$

KKT conditions for unconstrained optimal control

Let $(\mathbf{x}^*, \mathbf{u}^*)$ be a regular point for the dynamics constraints and an optimal (state-input) trajectory. Then there exists $\boldsymbol{\lambda}^*$ such that $\nabla \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*) = 0$.

Let us explicitly write condition $\nabla_{(1,2)} \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*) = 0$

$$\nabla_{(1,2)} \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*) = \begin{bmatrix} \nabla_1 \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*) \\ \nabla_2 \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*) \end{bmatrix} = 0$$

Since $\mathcal{L}(\mathbf{x}, \mathbf{u}, \boldsymbol{\lambda}) = \sum_{t=0}^{T} \mathcal{L}_t(x_t, u_t, \boldsymbol{\lambda})$, we can exploit this sparsity and write

$$\nabla_2 \mathcal{L}_0(x_0, u_0, \boldsymbol{\lambda}) = 0 \qquad \nabla_2 \ell(x_0, u_0) + \nabla_2 f_0(x_0, u_0) \lambda_1 = 0$$

$$\begin{bmatrix} \nabla_1 \mathcal{L}_t(x_t, u_t, \boldsymbol{\lambda}) \\ \nabla_2 \mathcal{L}_t(x_t, u_t, \boldsymbol{\lambda}) \end{bmatrix} = 0 \qquad \begin{bmatrix} \nabla_1 \ell(x_t, u_t) + \nabla_1 f_t(x_t, u_t) \lambda_{t+1} - \lambda_t \\ \nabla_2 \ell(x_t, u_t) + \nabla_2 f_t(x_t, u_t) \lambda_{t+1} \end{bmatrix} = 0 \qquad t = 1, \dots, T-1$$

$$\nabla_1 \mathcal{L}_T(x_T, \boldsymbol{\lambda}) = 0 \qquad \nabla \ell_T(z_T) - \lambda_T = 0$$

for $(x_t, u_t, \lambda_t) = (x_t^*, u_t^*, \lambda_t^*)$ for all $t = 0, \dots, T$.

KKT conditions for constrained optimal control

Recall KKT conditions for unconstrained optimal control

Let $(\mathbf{x}^*, \mathbf{u}^*)$ be a regular point for the dynamics constraints and an optimal (state-input) trajectory. Then there exists $\boldsymbol{\lambda}^*$ such that $\nabla \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*) = 0$, i.e.,

$$\nabla_{(1,2)} \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*) = \begin{bmatrix} \nabla_1 \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*) \\ \nabla_2 \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*) \end{bmatrix} = 0$$

and

$$h(\mathbf{x}^*, \mathbf{u}^*) = 0$$

namely, $\mathbf{x}^*, \mathbf{u}^*$ is a trajectory of the system, i.e., it satisfies the dynamics.

Discrete-time optimal control problem (constrained)

We look for a solution of the discrete-time optimal control problem

$$\min_{x_0 \in \mathbb{R}^n, \mathbf{x} \in \mathbb{R}^{nT}, \mathbf{u} \in \mathbb{R}^{mT}} \quad \sum_{t=0}^{T-1} \ell_t(x_t, u_t) + \ell_T(x_T) \tag{5a}$$

$$\begin{aligned} \text{subj.to } & x_{t+1} = f_t(x_t, u_t), \quad t = 0, \dots, T-1 \\ & r(x_0, x_T) = 0, \\ & g_t(x_t, u_t) \leq 0, \quad t = 0, \dots, T-1 \end{aligned} \tag{5b}$$

where

- $\ell_t : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is the so-called stage cost,
- $\ell_T : \mathbb{R}^n \to \mathbb{R}$ is the terminal cost,
- $r : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{p_0}$ identifies a *boundary constraint* on initial and final states,
- $g_t : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^p$ for each t identifies *point-wise constraints* on state and input at some time t.

KKT conditions for constrained optimal control

The Lagrangian function has the form

$$\begin{aligned}\mathcal{L}(\mathbf{x},\mathbf{u},\boldsymbol{\lambda},\boldsymbol{\mu}) &= \ell(\mathbf{x},\mathbf{u}) + \boldsymbol{\lambda}_d^\top h(\mathbf{x},\mathbf{u}) + \lambda_b^\top r(x_0,x_T) + \boldsymbol{\mu}^\top g(\mathbf{x},\mathbf{u}) \\ &= \underbrace{\sum_{t=0}^{T-1} \ell_t(x_t,u_t) + \ell_T(x_T)}_{\ell(\mathbf{x},\mathbf{u})} + \underbrace{\sum_{t=0}^{T-1} \lambda_{d,t+1}^\top (f_t(x_t,u_t) - x_{t+1})}_{\boldsymbol{\lambda}_d^\top h(\mathbf{x},\mathbf{u})} + \lambda_b^\top r(x_0,x_T) + \underbrace{\sum_{t=0}^{T-1} \mu_t^\top g_t(x_t,u_t)}_{\boldsymbol{\mu}^\top g(\mathbf{x},\mathbf{u})} \\ &= \sum_{t=0}^{T} \mathcal{L}_t(x_t,u_t,\boldsymbol{\lambda})\end{aligned}$$

where

$$\begin{aligned}\mathcal{L}_0(x_0,u_0,\boldsymbol{\lambda},\boldsymbol{\mu}) &= \ell_0(x_0,u_0) + \lambda_{d,1}^\top f(x_0,u_0) + \lambda_{b,0}^\top r_0(x_0) + +\mu_0^\top g_0(x_0,u_0) \\ \mathcal{L}_t(x_t,u_t,\boldsymbol{\lambda},\boldsymbol{\mu}) &= \ell_t(x_t,u_t) + \lambda_{t+1}^\top f(x_t,u_t) - \lambda_t x_t + \mu_t^\top g_t(x_t,u_t) \qquad t = 1,\dots,T-1 \\ \mathcal{L}_T(x_T,\boldsymbol{\lambda}) &= \ell_T(x_T) - \lambda_T^\top x_T + \lambda_{b,T}^\top r_T(x_T)\end{aligned}$$

and we assumed $r(x_0,x_T) = \mathrm{col}(r_0(x_0), r_T(x_T))$ and $\lambda_b = \mathrm{col}(\lambda_{b,0}, \lambda_{b,T})$.

KKT conditions for constrained optimal control

Let $(\mathbf{x}^*, \mathbf{u}^*)$ be a regular optimal (state-input) trajectory. Then there exist $\boldsymbol{\lambda}^*$ and $\boldsymbol{\mu}^* \geq 0$ such that $\nabla \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = 0$ + complementary slackness conditions.

Let us explicitly write condition $\nabla_{(1,2)} \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = 0$

$$\nabla_{(1,2)} \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \begin{bmatrix} \nabla_1 \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \\ \nabla_2 \mathcal{L}(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \end{bmatrix} = 0$$

Since $\mathcal{L}(\mathbf{x}, \mathbf{u}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \sum_{t=0}^{T} \mathcal{L}_t(x_t, u_t, \boldsymbol{\lambda})$, we can exploit this sparsity and write

$$\begin{bmatrix} \nabla_1 \mathcal{L}_0(x_0, u_0, \boldsymbol{\lambda}, \boldsymbol{\mu}) \\ \nabla_2 \mathcal{L}_0(x_0, u_0, \boldsymbol{\lambda}, \boldsymbol{\mu}) \end{bmatrix} = 0 \qquad \begin{bmatrix} \nabla_1 \ell(x_0, u_0) + \nabla_1 f_0(x_0, u_0) \lambda_1 + \nabla r_0(x_0) \lambda_{b,0} + \nabla_1 g_t(x_0, u_0) \mu_0 \\ \nabla_2 \ell(x_0, u_0) + \nabla_2 f_0(x_0, u_0) \lambda_1 + \nabla_2 g_t(x_0, u_0) \mu_0 \end{bmatrix} = 0$$

$$\begin{bmatrix} \nabla_1 \mathcal{L}_t(x_t, u_t, \boldsymbol{\lambda}) \\ \nabla_2 \mathcal{L}_t(x_t, u_t, \boldsymbol{\lambda}) \end{bmatrix} = 0 \qquad \begin{bmatrix} \nabla_1 \ell(x_t, u_t) + \nabla_1 f_t(x_t, u_t) \lambda_{t+1} - \lambda_t + \nabla_1 g_t(x_t, u_t) \mu_t \\ \nabla_2 \ell(x_t, u_t) + \nabla_2 f_t(x_t, u_t) \lambda_{t+1} + \nabla_2 g_t(x_t, u_t) \mu_t \end{bmatrix} = 0 \quad t = 1, \dots, T-1$$

$$\nabla_1 \mathcal{L}_T(x_T, \boldsymbol{\lambda}) = 0 \qquad \nabla \ell_T(x_T) - \lambda_T + \nabla r_T(x_T) \lambda_{b,T} = 0$$

for $(x_t, u_t, \lambda_t, \mu_t) = (x_t^*, u_t^*, \lambda_t^*, \mu_t^*)$.